GeoTruth AVM

A multi-modal property valuation engine for Indian housing finance

A concept for bridging the gap between statutory circle rates, portal listings and property-level evidence through spatial learning and confidence-led verification.



SUBMITTED BY

Utkarsh Verma

TRACK

AI-Powered Digital Property Valuation Engine

SUBMISSION DATE

12 November 2026

Concept proposal. Performance, latency and inspection-reduction figures are proposed targets, not measured results; they require validation in a controlled lender pilot.

01 / PROBLEM CONTEXT

Why existing automated valuation models struggle

Many automated valuation models are designed around relatively structured property records and standardized addresses. In affordable housing and Loan Against Property (LAP) markets across Indian Tier-2 and Tier-3 cities, three recurring data problems can weaken a valuation: price-source disagreement, limited visibility into the built structure, and inconsistent property identity.

01

Price-signal gap

Circle rates are statutory reference values, while portal listings are asking prices. Either can diverge from a completed transaction. Treating one source as ground truth can bias the estimate.

02

Structure-visibility gap

Overhead imagery can reveal a footprint, but not reliably establish internal condition, dampness, structural distress or every unapproved floor addition.

03

Identity and area gap

Landmark-based addresses, vernacular naming and mixed carpet / built-up / super-built-up measurements complicate entity matching and like-for-like comparisons.

THE PROPOSED RESPONSE

GeoTruth connects three mechanisms in one lender-facing workflow. Each mechanism addresses a different source of valuation uncertainty.

A

Spread-anchored estimate

Use circle-rate and de-duplicated portal signals as reference anchors, with transaction evidence where available.

→

B

Spatial and visual context

Add H3-neighbourhood relationships, accessibility, building-footprint cues and optional property-level imagery.

→

C

Confidence-led verification

Route uncertain cases to a guided borrower capture or a field engineer, rather than treating every output as equally reliable.

DESIGN PRINCIPLE

A valuation should be accompanied by its evidence, uncertainty range and verification path. GeoTruth is proposed as a decision-support layer for lender workflows, not as an autonomous substitute for credit or legal review.

02 / MARKET SCOPING

Segment the market before scaling

A single model should not assume that a registered high-rise apartment, an independent house and a mixed-use shop share the same evidence profile. The rollout below groups markets by data maturity and property type.

PHASE 1

MONTHS 1–6

DATA-DENSE METROS AND TIER-1.5 HUBS

GEOGRAPHY

Mumbai MMR, Delhi-NCR, Bengaluru, Pune, Hyderabad, Chennai, Lucknow, Jaipur and Indore.

ASSET TYPES

Residential: RERA-registered apartments and gated builder floors. Commercial: standardized office units and high-street retail.

SCALING MECHANISM

Start with a tabular and spatial ensemble trained on IGR registration deeds, historical lender valuations and RERA project filings.

PHASE 2

MONTHS 6–12

TIER-2 AND TIER-3 LAP GROWTH CENTRES

GEOGRAPHY

35+ industrial and trading cities, including Surat, Nagpur, Coimbatore, Kanpur, Bhopal, Ludhiana and Mysuru.

ASSET TYPES

Residential: independent houses and freehold plots. Commercial: mixed-use ground-floor shops and light warehouses.

SCALING MECHANISM

Use spatial transfer learning with state circle-rate anchors, Sentinel-2 shadow-based height cues and guided video verification for independent structures.

PHASE 3

MONTHS 12–18

PERI-URBAN AND EMERGING TOWNS

GEOGRAPHY

Tier-3/4 municipalities and highway growth corridors across 18+ states.

ASSET TYPES

Self-constructed houses, semi-commercial properties and arterial-road assets.

SCALING MECHANISM

Combine guided telemetry capture with VIIRS night-time-light context and India Post DIGIPIN spatial indexing.

Rollout gate: Move each cohort forward only when address resolution, comparable-sale coverage, data rights and out-of-time validation meet agreed thresholds. City lists and phase timings are planning assumptions, not committed coverage.

03 / MULTI-MODAL DATA FUSION

Build a consistent property evidence layer

GeoTruth proposes a unified spatial index that links structured records, property attributes, satellite / geospatial evidence and neighbourhood-level market signals. The design uses Uber H3 Resolution 9 cells (approximately 0.10 km² per cell, depending on location) cross-referenced with India Post DIGIPIN (a 4 m × 4 m grid).

DATA CLASS	PROPOSED INPUTS	VALUATION SIGNALS
Statutory and transaction	State IGR e-search portals; stamp-duty and circle-rate tables; state RERA registries; historical lender valuation records; CERSAI records.	Registered consideration and date; circle-rate reference; RERA carpet area; approved floor count; completion history.
Property and micro-attributes	Municipal property-tax rolls; borrower application; licensed aggregator APIs (99acres, MagicBricks, Housing.com, Zapkey, PropStack).	Asking-price distribution; structure age; floor number; parking; corner-plot status; road access and commercial frontage.
Satellite and geospatial	Sentinel-2 and ISRO Bhuvan; Google Open Buildings / Microsoft building footprints; OpenStreetMap; PM Gati Shakti layers.	Footprint area; candidate height / floor-count cues; travel-time isochrones; proximity to services and mapped risk areas.
Macro and market trends	NHB RESIDEX; VIIRS night-time-light radiance; metro DPRs; highway announcements; commercial points of interest.	Neighbourhood price momentum; infrastructure signals; commercialisation trends and local economic activity proxies.

HARMONISATION PIPELINE

01 **Normalize area**

Convert legacy super-built-up / built-up measures to a common carpet-area basis using documented, property-specific factors. The source proposal uses 0.70–0.76 as a typical apartment loading range; this must be calibrated by asset and locality.

02 **Screen transaction outliers**

Flag family transfers, distress auctions and non-market deeds; apply robust statistical screening such as IQR only with review rules that avoid removing genuine market events.

03 **Resolve address entities**

Use IndicBERT-style text matching and phonetic normalization to map vernacular variants (for example, “Mandir Marg” and “Temple Road”) to candidate DIGIPIN and H3 identifiers.

04 **Derive spatial features**

Join imagery, road access, amenities and solar-shadow geometry. A candidate height estimate is $H = L_{shadow} \times \tan(\theta)$, with $N_{floors} \approx \text{round}(H / 3.1\ \text{m})$; validate timestamp, solar angle, obstruction and roof geometry before using it.

05 **Maintain a time-aware store**

Use PostGIS to join rolling 6-, 12- and 36-month market signals and infrastructure-distance buffers to each H3 cell.

Data control: Source access, licensing, timestamp quality, geocoding confidence and provenance should be recorded for every feature. Satellite-derived cues should not be treated as proof of internal condition or legal approval.

04 / HYBRID AI / ML ARCHITECTURE

Estimate value, then quantify uncertainty

The proposed model is a constrained multi-branch ensemble. It combines structured property attributes, local transaction relationships and visual / telemetry condition cues. Circle rates and portal listings are reference anchors—not guaranteed lower and upper bounds in every locality.

SPREAD-ANCHORED VALUATION FORMULATION

$$\text{FMV per sq. ft.} = [P_{circle} + \alpha(P_{ask} - P_{circle}) + \Delta_{spatial}] \times \gamma_{CV}$$

α is the learned position within the observed price spread (proposed range 0–1.25); Δspatial is the neighbourhood adjustment; γCV is a visual / condition multiplier (proposed range 0.80–1.15). Because α can exceed 1 and adjustments are applied, the final estimate is not mathematically bounded by the two anchors.

01 Tabular hedonic and spread predictor

LightGBM / CatBoost uses 65+ structured features such as carpet area, building age, floor level, road width, NHB RESIDEX trend and commercial frontage. Monotonic constraints should be applied only where supported by local evidence and policy.

02 Spatial graph model

GraphSAGE over H3 cells represents a property and candidate nearby registered sales. The proposal specifies 15 nearest sales within 1.5 km, with edge weights reflecting road distance, transaction age and property-type similarity; sparse areas need a fallback strategy.

03 Visual and telemetry condition scorer

ResNet-50 / YOLOv8 is proposed for roof-footprint and road-access cues; guided smartphone frames may add visible-condition signals. The model cannot infer concealed structural defects from a short walkthrough with certainty.

UNCERTAINTY AND PERFORMANCE TARGETS

A split-conformal layer can construct prediction intervals; 90% coverage is a validation target under suitable exchangeability assumptions, not an unconditional guarantee. The source's routing score is C = max[0, 1 − (FMVhigh − FMVlow) / FMVpoint] × 100%; calibrate it against observed errors.

METRIC	LEGACY BASELINE IN PROPOSAL	GEOTRUTH PROPOSED TARGET
MdAPE	14.5%–18.0%	≤ 6.2% Tier-1 flats; ≤ 8.5% Tier-2 independent / LAP
PPE10	52%–58%	≥ 82% within ±10% of reference value
PICP-90	Not calibrated	≥ 90% empirical interval coverage on held-out data

Evidence status: The figures above are planning targets copied from the proposal. They are not measured GeoTruth results. Baseline values and target thresholds should be sourced, defined and tested using the same property cohorts and reference-value methodology.

05 / EXPLAINABLE AI AND AUDIT DOSSIER

Make every valuation reviewable

A lender-facing valuation should expose the inputs, adjustments, confidence range and evidence used. The proposed Valuation Audit Dossier is intended to help credit underwriters and internal auditors understand why the model produced a particular estimate.

ILLUSTRATIVE LUCKNOW RESIDENTIAL EXAMPLE

Hypothetical 1,100 sq. ft. unit in Gomti Nagar Extension, Sector 4. The numbers below illustrate the proposed explanation format; they are not a verified appraisal or a live property record.

₹4,450	₹4,750	₹52.25 lakh
Starting local benchmark per sq. ft.	Illustrative final FMV per sq. ft.	Illustrative unit value 1,100 sq. ft.

FACTOR	OBSERVED / ASSUMED INPUT	MODEL ADJUSTMENT	UNIT IMPACT
Base locality benchmark	Circle rate ₹3,400; portal ask ₹5,600; local anchor ₹4,450.	Starting point	₹48,95,000
Infrastructure and transit	420 m walk to operational metro corridor; 18 m arterial road.	+₹340 / sq. ft.	+₹3,74,000
Plot geometry	Corner plot; two sides open; two floors assumed verified.	+₹190 / sq. ft.	+₹2,09,000
Age and condition	11 years; municipal record and interior telemetry assumed.	−₹230 / sq. ft.	−₹2,53,000
FINAL ILLUSTRATIVE FMV	₹4,450 + ₹340 + ₹190 − ₹230 = ₹4,750 / sq. ft.	₹4,750 / sq. ft.	₹52,25,000

VALUE RANGE

90% interval shown in the source proposal: ₹49.6L–₹54.8L. The interval must be calibrated and monitored on held-out data.

CONFIDENCE

Source example: 89.4%. Treat the displayed score as a model-routing indicator until calibrated against observed error.

FORCED-SALE VIEW

Source example applies an 85% factor, yielding approximately ₹44.41L. The haircut must be lender- and market-policy governed.

Audit artefacts to retain: SHAP feature contribution summary; five comparable-sale references and map; source and timestamp provenance; imagery / telemetry references; model version, calibration set and reviewer actions.

06 / SYSTEM ARCHITECTURE AND DECISION SUPPORT

Route cases by confidence and lending risk

GeoTruth is proposed as a cloud-native, API-first service connected to a lender's Loan Origination System (LOS). The workflow moves from evidence collection to valuation, confidence routing and an auditable lender response.

01 INGEST
IGR, circle rates, RERA, portals, satellite / GIS and borrower inputs.

02 ESTIMATE
LightGBM / CatBoost, GraphSAGE, visual cues and conformal interval.

03 ROUTE
Green: desktop review. Amber: guided capture. Red: field inspection.

04 DELIVER
FMV, forced-sale view, LTV inputs, risk flags and audit dossier to LOS.

CONFIDENCE ROUTING AND LENDER ACTION

CHANNEL	TRIGGER / PROPERTY PROFILE	VERIFICATION AND TARGET TAT	PROPOSED LENDING TREATMENT
GREEN	C ≥ 85%; loan ≤ ₹75 lakh. High-liquidity cluster (proposal threshold: >25 registered sales / year in H3 cell).	Desktop AVM; target under 8 seconds. No additional borrower capture in the standard path.	Indicative ceiling: up to 75%–80% home-loan LTV or 65% LAP, subject to lender policy and applicable rules. Standard pricing only after credit checks.
AMBER	60% ≤ C < 85%; Tier-2 independent property / LAP. Moderate liquidity (8–25 sales / year).	45-second guided smartphone capture with GPS / IMU and visual review; target under 3 minutes end-to-end.	Use the lower confidence bound as a conservative input; proposed 60%–70% LTV range. Consider a 15–35 bps risk premium only after policy calibration.
RED	C < 60%, loan > ₹1.5 crore, title / GIS flag, low liquidity or non-standard asset.	Targeted engineer inspection and legal / title review; proposal TAT 24–36 hours.	Proposed 45%–55% cap against forced-sale value or manual credit sign-off. No automated approval.

Important: Thresholds, LTV ranges, risk premiums and turnaround times are proposed operating parameters from the source document. The stated 65% / 25% / 10% Green-Amber-Red mix and potential inspection reduction of up to 75% are planning hypotheses, not observed results. Validate all figures against the lender's credit policy, regulatory requirements, fraud controls and service-level capacity.

Design intent: Automate routine, high-confidence collateral review; collect additional evidence when uncertainty is material; preserve human review for low-confidence, high-value or flagged cases.

07 / GOVERNANCE, TESTING AND NEXT STEPS

Validate before scaling

The proposal should be evaluated as a lender decision-support system, not only as a price-prediction model. Model performance, confidence calibration, operational speed and inspection savings should be measured independently.

PROPOSED VALIDATION PROTOCOL

TEST	PROPOSED METHOD	WHAT IT SHOULD ESTABLISH
Out-of-time backtesting	Train on the prior 36 months; test on the most recent 6 months.	Whether performance holds on later transactions, without future-data leakage.
Spatial block cross-validation	Hold out entire H3 clusters rather than randomly splitting nearby properties.	Whether the model generalises to unseen localities instead of memorising micro-markets.
Reference comparison	Compare outputs with historical Hero Housing Finance engineer valuations and registered IGR sale deeds, where matched and appropriate.	Error distribution, segment bias and the reliability of each reference source.
Conformal calibration	Measure prediction-interval coverage and width by city, property type and time period.	Whether stated 90% coverage is achieved on relevant held-out cohorts.
Operational pilot	Track capture completion, red-flag rates, review time, inspection dispatches and lender overrides.	Whether automation reduces effort without weakening auditability or risk controls.

KEY CONTROLS BEFORE PRODUCTION

- Evidence quality: log source, timestamp, licensing basis, geocoding match and missingness for every input.
- Model limits: do not equate roof-shadow height cues or short video scans with proof of floor legality or structural soundness.
- Privacy and consent: define borrower consent, retention, access control and deletion rules for location and video data.
- Human oversight: preserve engineer / legal review for low-confidence, title-flagged, unusual or high-value cases.
- Drift monitoring: re-check calibration and segment performance when transaction mix, geography or data-source coverage changes.

REFERENCES CITED IN THE SOURCE PROPOSAL

1. Ministry of Housing and Urban Affairs (2016). Real Estate (Regulation and Development) Act, 2016 — definition of carpet area, Section 2(k).

2. Department of Posts, Ministry of Communications, and IIT Hyderabad (2024). DIGIPIN: National Level Addressing Grid Specification for Geo-Coded Digital Addressing in India.

3. National Remote Sensing Centre / ISRO (2024). Bhuvan Geo-Platform and urban master-plan geospatial layers.

4. National Housing Bank (2024). NHB RESIDEX: Residential Property Price Indices Across 50 Cities in India — Technical Methodology.

5. Angelopoulos, A. N., and Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification.

6. Lundberg, S. M., and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions (SHAP). NeurIPS.

7. RBI / NHB (2021/2023). Master Direction — Non-Banking Financial Company — Housing Finance Company (Reserve Bank) Directions.

APPENDIX A / REQUIREMENT TRACEABILITY

How this proposal maps to Problem Statement 1

The challenge brief requires all six items on its "What you need to do" page. The table below maps each item to the submitted design and separates design coverage from evidence that would only be available after a pilot.

CHALLENGE REQUIREMENT	WHERE ANSWERED	WHAT THE PROPOSAL PROVIDES	WHAT STILL NEEDS VALIDATION
1. Scope the market	**Page 3**	Residential and commercial asset types; initial city clusters; three-phase expansion across data-dense, Tier-2/3 and emerging markets.	Confirm usable records, local coverage and partner permissions city by city.
2. Combine data sources	**Page 4; Appendix B**	Structured records, listings, remote sensing, geospatial and market signals; area normalization, outlier review, entity resolution and feature-store joins.	Confirm access rights, update frequency, quality and lawful reuse for each source.
3. Build the AI/ML model	**Page 5; protocol below**	Tabular model, spatial graph model, visual/telemetry branch, prediction intervals and proposed accuracy metrics.	Run a matched comparison against an incumbent AVM and reproducible baselines; publish results by segment.
4. Make it explainable	**Page 6**	Illustrative FMV, factor-level adjustments, interval, routing score and audit artefacts.	Generate these artefacts from real model runs and validate calibration and reviewer usefulness.
5. Automate and scale	**Pages 3 and 7**	Four-stage workflow, confidence-based routing, proposed turnaround times and phased geographic rollout.	Measure latency, completion rate, overrides, operating cost and inspection savings in a controlled pilot.
6. Enable lending decisions	**Page 7; regulatory note on page 10**	FMV/forced-sale view, conservative value routing, LTV inputs, risk flags and escalation to human review.	Map thresholds to the lender's current product policy and applicable RBI directions before deployment.

Submission interpretation

This is a concept and solution-design submission. The brief says a prototype or architecture diagram is welcome; it does not make a prototype mandatory. The current PDF describes the data layer (page 4), model (page 5), valuation output (page 6) and lender workflow (page 7). It does not claim that a production model or live Hero integration already exists.

Accuracy: the benchmark plan that makes the claim testable

The phrase "significantly higher accuracy than existing AVMs" should be treated as a testable objective, not a result. Use the same held-out properties, reference values, time window and geography for every comparator.

STEP	PROTOCOL	REPORT
1. Define the test set	Use verified registered sales and/or lender engineer valuations after matching, deduplication and reference-quality review. Preserve a time-based holdout and spatially held-out H3 clusters.	Sample size, coverage, time period, city, asset type, exclusions and reference-source quality.
2. Run fair comparators	Compare: incumbent lender AVM where supplied; a documented conventional hedonic / tabular baseline; a locality-and-asset median baseline; and the full GeoTruth model. Keep training data and test cases matched.	Ablation results for tabular-only, spatial branch, visual branch and full ensemble. State clearly if no incumbent AVM was made available.
3. Measure outcomes	Report MdAPE, MAE/RMSE, PPE10 and 90% interval coverage and width. Break out results by property type, city / tier, price band and data-density group.	Point estimates, sample counts and uncertainty intervals (for example, paired bootstrap confidence intervals).
4. Set a claim threshold	Claim an improvement only where the matched holdout shows a material and statistically supported uplift without unacceptable segment-level degradation or poor calibration.	A pre-declared pass criterion and an error-analysis summary, including failure cases.

Baseline caution: The 14.5%–18.0% MdAPE and 52%–58% PPE10 figures shown as "legacy baseline in proposal" on page 5 are provisional figures carried forward from the initial concept draft. No source or matched benchmark dataset is supplied for them in this submission. They must not be presented as verified, market-wide performance statistics.

APPENDIX B / DATA PROVENANCE AND TECHNICAL CONTROLS

Source register, access conditions and current rules

The challenge asks teams to cite the data and sources they use. These are proposed source families, not a claim that GeoTruth already has agreements or live access. Access, reuse rights, coverage and metadata must be verified before implementation.

SOURCE FAMILY	PROPOSED USE AND ACCESS CONDITION	REFERENCE
State IGR, stamp-duty, circle-rate, RERA and municipal records	Registered consideration, statutory anchors, project / building attributes and tax assessments. Portals, formats and disclosure rules differ by State/UT and city; confirm permissions, completeness and any bulk/API access.	State / UT registration and RERA authorities; municipal bodies (state-specific).
Property listings and lender / registry records	99acres, MagicBricks, Housing.com, Zapkey and PropStack are proposed commercial feeds: use only under provider-approved licences or contracts; asking prices are not completed sale prices. CERSAI, where the lender is entitled to access it, is a collateral charge / encumbrance signal—not a substitute for transaction-price evidence.	Provider contracts and lender-authorised access; confirm legal basis before ingestion.
NHB RESIDEX	Use as a city / market trend feature, not as a parcel-level comparable. NHB states that the FY 2024-25 base-year series applies from the quarter ended March 2026 and coverage is 50 cities.	NHB RESIDEX official information
Satellite imagery and building footprints	Sentinel-2 visible / near-infrared bands are 10 m; other bands are 20 m or 60 m. Treat Sentinel-2 as neighbourhood / land-cover context, not reliable stand-alone evidence of an individual roof shadow or floor count. Building-level height cues need suitable higher-resolution licensed imagery and validation. Bhuvan products vary by layer and metadata. Google Open Buildings V3 and Microsoft Global ML Building Footprints are candidate footprint layers; verify India coverage, imagery vintage, positional quality and licence before relying on them.	Copernicus Sentinel-2 specs NRSC / Bhuvan services Google Open Buildings V3 Microsoft Global ML Building Footprints
PM GatiShakti / UGI and project documents	Use selected approved non-sensitive infrastructure layers and query outputs available through the public interface. Do not assume the full National Master Plan database is openly downloadable or unrestricted. For metro DPRs and highway announcements outside the interface, cite dated official project documents and distinguish announced, under-construction and operational assets.	PIB: PM GatiShakti Public / UGI
VIIRS nighttime lights	NASA Black Marble VNP46A2 provides a 500 m nighttime-lights product. Use as a broad neighbourhood / economic-activity proxy only—not as a property-level feature or direct price measure.	NASA VIIRS Black Marble product
DIGIPIN, H3 and OpenStreetMap	DIGIPIN is a geocoded address grid (finest level approximately 3.8 m); it is not proof of ownership or cadastral boundaries. H3 resolution 9 has an average hexagon area of about 0.1053 km². OpenStreetMap is ODbL data: provide attribution and observe its licence obligations.	India Post DIGIPIN technical document Uber H3 resolution table OpenStreetMap copyright / licence

Regulatory note — replace the older dated reference

The current reference for Housing Finance Companies is the Reserve Bank of India (Housing Finance Companies) Directions, 2025, updated as on 1 October 2026. For HFC housing loans to individuals, paragraph 58 sets maximum LTV ratios of 90% for loans up to ₹30 lakh, 80% for loans above ₹30 lakh and up to ₹75 lakh, and 75% for loans above ₹75 lakh. The denominator is the realizable value of the residential property mortgaged to the HFC. These are regulatory ceilings, not recommended loan offers; a lender may apply lower limits.

The proposal's 65% LAP figure is a proposed risk-policy parameter, not a universal statutory LAP ceiling. The RBI Directions define housing finance by specified residential purposes; loans secured by a mortgage for other purposes are treated as non-housing loans under paragraph 10(8). Do not assume the home-loan LTV caps govern every LAP product. Configure LAP separately against the lender's current product policy and legal review. The 2025 Directions replace the older 2021/2023 reference wording in the initial draft.

RBI — Housing Finance Companies Directions, 2025 (updated 1 October 2026)

Final implementation gates

Before any pilot: (1) document the licence / permission and refresh cadence for every input; (2) record provenance, timestamps, geocoding match quality and missingness; (3) obtain borrower consent and define retention / deletion controls for location and video; (4) keep human review for title, legal, low-confidence and exceptional cases; and (5) validate model drift, calibration and segment bias. Source URLs and access conditions checked on 12 November 2026.